Intelligent RFID Based Library Management System

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***Abstract*-The term RFID stands for Frequency Identification Technology that may be a distinctive technology that uses magnetism fields to keep track of things, products or objects. Intelligent-RFID Technology in libraries helps us to analyze and predict the demand of books, books that are less utilized, count of books issued in each department, fine calculation, book enquiry, percentage ratio of total number of books in the library to total books issued, thus imparting intelligence to libraries. The process of automating the library tasks addresses the issues relating to book-searching, and tracking, thereby ensuring the safety and security of library materials. This paper proposes an Intelligent and Automated RFID based library management system along with user interface for different types that enables the quick flow of books. A PyQt web portal is designed for creating user interface to perform the desired library tasks and predictive analysis. In addition, data processing is performed to analyze various parameters required to produce statistical analysis on extensively used or less used books, which helps us to protect or discard the books from time to time and prevent the library assets from termite attacks. Fine calculation helps to calculate the fine and send text information to the users in case of expiry of due dates or lost books from some of the borrowers.**

***Keywords* — Radio Frequency Identification (RFID), PyQt, Library management, Fine Calculation, Predictive analysis.**

I. INTRODUCTION

RFID technology is now being adopted widely in various large-scale applications and big retail chains across the world to efficiently track assets, manage stocks, and control quality processes. Bar-coding technology which is now being replaced by RFID Technology offers significant benefits. RFID Tags do not require line-of-sight scanning and can read in a bulk. RFID Tags can be read at much greater distances (up to 300ft) in certain applications. It also offers high durability, increased data capture, security, and improved productivity. This also has enhanced its utility when IoT and Sensor technologies have strongly deployed in every sectors.

Using such RFID Technology in Library management helps in quick issues, return, and re-issue process of books. This technology helps in the direct transaction of information from the RFID Tags attached to the books, accessing through the reader and to the PC (database system). The updating and validation process in the main database reduces the theft of assets. Each RFID tags programmed with a unique inventory code, which gets scanned when zapped against the RFID Reader.

When an RFID tag is scanned by the reader, the reader recognizes the unique inventory code and immediately updates the account of the user. The RFID tags store book numbers (ID), Author info, etc.., however, Bar-code Technology is limited to hold very little primary step which involves procedures and techniques to convert physically accessed data into an understandable format Normally, many a time, real world data is often inconsistent, incomplete and lacks certain behavior and trends. In this work a data processing technique is used by database driven applications attached with RFID technology that provides an effective and consistent analysis over the assets monitored in library management system. In this proposed model, data processing is carried out in order to analyze data and obtain useful library information for classification and prediction as an intelligent system along with the RFID hardware module.

The remaining work is discussed in the following manner. Section II provides a brief overview of related works in this field. Section III describes the existing model as well as the suggested system design. Section IV explains the proposed methodology and techniques, whereas Section V examines the implementation, execution and outcomes. Sections VI and VII highlights the conclusions and future scope of the study, respectively.

II. RELATED WORKS

This section discusses various works related to RFID Technology in Library Management Systems and data analytics.

RFID based library management system is projected in paper [1] uses a high-frequency DLP (Data Loss Prevention) RFID Read/Write that ensures knowledge security and safety. The computer code used by the library management system is MATLAB and MYSQL to enhance system performance and potency. The system was successful, enforced, and tested. In [2] the work involves in resolving issues relating to time, arrangement, and searching of library books. To overcome the problems, library information like the range of books, Name, Author, etc., are digitized and kept over a secure network. RFID Technology is enforced for easy, fewer confusing transactions and has replaced the Bar-code Technology. RFID-based Library Management System that permits quick dealing flow, creating an issue/return of books easier with no human intervention for manual book- keeping. In [3] This method conjointly calculates the corresponding fine related to the period of the absence of the book from the library information. In [4] RFID Technology is adopted to contour workflow in areas of self-service, book exchange, shelf, and inventory management. The paper conjointly discusses sensible problems associated with RFID operating in a library setting and problems regarding maintenance, computer code compatibility, privacy, and standardization (NISO). In [5] Additionally, the quick flow of books, self-check solutions, scanning multiple books at the same time reduces the time interval required to check out from the library. The protection bit is fittingly set, thereby preventing the thievery or burglary of the library assets. A library management system to spot and find remotely placed books that help the speedy inventory method thereby enhances client expertise using RFID system. In [6] enhanced circulation and optimization techniques are used for the allocation of labor and money resources. Paper [7] stresses on security features of the library materials and implementation methodologies. In [8] a robot-based library management system is proposed, that uses UHF (Ultra

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High Frequency), mobile reader, to extend the potency of library materials management. The book ID is employed to store the distinctive serial range on ultrahigh-frequency RFID tags. Android Studio IDE is employed to develop an application computer code. This makes the library system a lot sturdy, robust and richer in operation.

III. Existing Model And Proposed System Design

A. Concept:

RFID Technology has supplanted the standard Bar-coding framework these days. Below figure 1 portrays the bit-by-bit measure associated with building the proposed model. The RFID Library Management System comprises an RFID tag, an RFID reader, and the Raspberry Pi. Each book in the library is embedded with an RFID tag in it. Each RFID tag is customized with a distinctive inventory number. The RFID tags contain a book name, Author information, etc. The RFID module that comprises the RFID reader discharges radio waves that trigger the microchip engraved on the RFID tag. The distinctive RFID number is passed on to the reader which is displayed on the LCD and on the Desktop App, made utilizing PyQt GUI. The required process like the issue/return of the book is immediately handled by the Raspberry Pi and is updated onto the library's principle Database i.e., Fire Base. Thus, issues like waiting time for the submit/issue process and searching becomes simple.

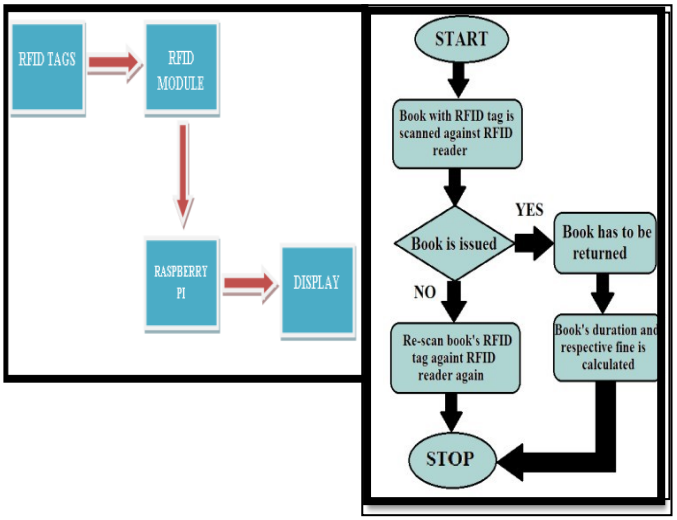

Figure 1: RFID based Library books management Model and System flow

The above figure 1 also depicts the system flow process i.e., working of RFID-Based Library Management System. Here, the book with the RFID tag embedded with it, is scanned by the RFID reader. The micro controller i.e., Raspberry Pi is programmed to determine the different status of the book from the database like, availability, category, etc. Based on the access request and transactions the status of the book gets updated in the database and accordingly the information is transferred to the respective authorities and decision support system, like mobile messaging, library personal, fine calculation system, etc.

In addition to these features, the proposed model also classifies information based on pre-processed data and gives graphical results. These results help to analyze the count of books issued, number of books present in the library, count of department wise books, percentage ratio of total books in the library to the issued books. Once the raw data is pre-processed, normalized, cleaned and analyzed the data is then fed to a classification regression algorithm to classify the books category based on ISBN, ISSN scientific name of the book, etc., and also to predict the number of books that must be discarded, based on usage and access patterns.

In addition, prediction is also made as to which book belongs to which department using the Decision Tree. This feature reduces confusion when redundancy of book exists between different departments.

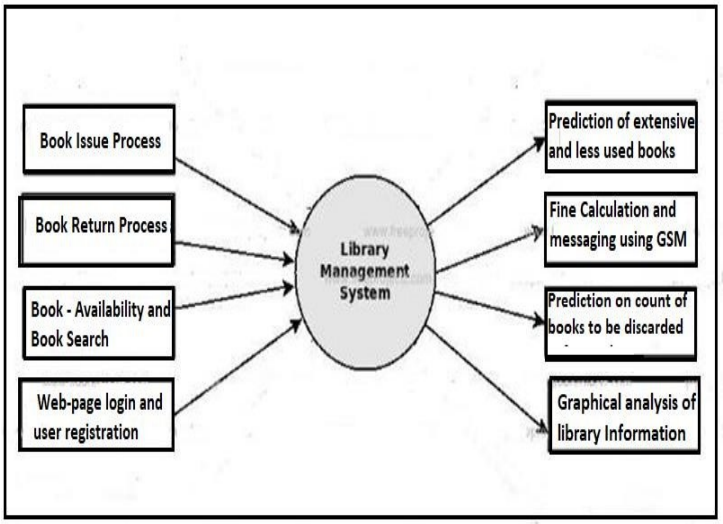

Figure 2: Tasks performed by the proposed model

The Intelligent RFID Library Management System consists of various smart features such as the Book issue, Book return, Book availability and Book search processes which helps the user to keep track of the Books. Moreover, it has a specially designed web-page login so that the user can register his/her details. To evolve the system more intelligently, it can predict extensively used books as well as books that are least in demand.

Fine Calculation makes it easier to determine the fine in case of late submission and the placement of the respective books in the library. Therefore, the prediction of the least used books could be discarded. Lastly, a final graphical analysis of the data sets will be done to procure the information of the library.

B. Hardware Components Used:

The system consists of the subsequent hardware components:

1)Raspberry Pi: The Raspberry Pi is a microcontroller that has high computing power with necessary application accessories which will be controlled and tailored for specific user applications and tasks.**2) RFID Tags**: The frequency Identification (RFID)tags are tiny gadgets that use low-power magnetic attraction waves to induce, store, and send information to the RFID interrogators or readers among a definite range. **3)RFID Module**: The RFID reader or interrogator could be a tiny device consisting of an associate imprinted reception equipment (antenna), a frequency module, and an impressive unit. The EM waves are emitted from the imprinted antenna of the reader to trigger the tag's chip that permits the exchange of digital information between the reader and tag. It's additionally referred to as a transceiver.**4) LCD display**: LCD display is used to display desired information. In this case, we use LCD to display the RFID unique inventory number.

C. Software Used:

The system consists of the subsequent software involvements:

1)Raspbian OS: An operating system is the arrangement of basic programs and utilities which permit the Raspberry Pi to run in an

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appropriate user-friendly manner. The Raspbian operating system is productively advanced to utilize the Raspberry Pi. Raspbian has GUI which incorporates Python programming, office, games, and so forth. **2)Python (Spyder)**: Python is a high-level programming language that uses objects and classes in its programming style. Its high-level underlying data structures, conjoined with dynamic typing and dynamic composing makes it appealing for Quick Application Development as well as ease in learning. It also has a dynamic semantics. Python language is straight forward, simple to get familiar with the linguistic structure, and accentuates intelligibility and readability.

IV. Proposed Methodology And Prototype System

All the modules and proposed intelligence RFID library system is described in this section.

Some of the intelligence and statistical analysis are included in order to get the benefit from the electronic access and data catalog created based on the usage pattern. In this model, each book is individually distinguished by means of the RFID labels or tags appended to it and communication is established remotely between tags and readers. An RFID reader or interrogator would be placed in the library work area premises where in one should zap the book's RFID tag against the reader to perform the specific desired tasks like borrow/return. Information regarding the availability of a book can also be acquired by the users remotely which is a substitute for manual book searching. Thereby reducing a great deal of time and empowers proficient queue management. As activities on the set tagged resources are being recorded and tracked, information can be put to use accordingly to add statistical analysis and intelligence to the system. The below explained steps give us a clear idea of the working of the proposed model.

A. Creating Database:

A G-mail account has been set-up through which one can access the cloud named Firebase. The Firebase here acts as a Database. This Database comprises details such as the Book Cards, Books, Scanned RFID Cards, Student Cards. There by enabling the librarian to keep track of books. The main contents of the database consist of sub details such as registered book cards, registered books, scanned RFID card numbers, and student cards registered. The database also comprises of student mail ID's, student registration numbers and book issue and return dates.

B. Procedure and Development Platform with Analytics

System uses the Spyder as an open-source cross-platform Integrated development environment (IDE) for scientific programming or coding in Python language. After Spyder is launched, Compile and Run the developed code. The corresponding PyQt code is dumped and then compiled on Spyder platform. The PyQt code results in a pop–up Desktop app designed for various user operations like borrow, return etc. The library desktop app is developed, where each member of the library can access his/her account and view personal transaction details, and know about the book availability. Figure 3 depicts the functional view of the webpage created. This interface created reduces the time taken for transactions and standing in long queues as data is automatically uploaded onto the database immediately. As the internet is widely available now a days, the desktop app can be reached to a large number of users, irrespective of variation in devices.

Fine calculation module enables to calculate the respective fine associated with a particular book post the deadline of seven days.

Above steps and procedures proposed in the system using Spyder, PyQt, analytical modules and hardware system is given in the Pseudo code shown below.

Pseudocode:

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- Run the PyQt code on Spyder Platform – This displays Desktop App created for user interface.
- Read the RFID tag embedded on the book with RFID Reader.
- Display of unique inventory number – LCD Screen.
- Unique Inventory code is displayed on the PyQt Desktop app.
- Desired tasks such as Issue, Borrow, Return etc. is carried out via the User Interface (UI).
- During return of a book, fine associated is calculated
- Activities performed are immediately updated on to the Database (Firebase).
- To find extensively used or less used book, run the python code developed on Spyder platform.
- Analytical modules are generated using machine learning algorithms for classification and regression with the help of usage pattern datasets.
- The respective results obtained are displayed on the screen.

}

The Desktop app created using PyQt GUI acts as an interface between the user and the library data. It allows the user to perform actions like borrow, book search and availability check and return process in a much simple and less time-consuming way. Once the RFID cards have been scanned it is displayed in the Card Number display box of the w, post which the desired above-mentioned tasks can be carried out efficiently. It is a user-friendly interface with a lot of advantages. A functional view of the webpage is depicted in the below figure 3.

Fig 3: Functional View of Desktop App using PyQt GUI

In the procedure, the analytics involves the following i.e., the maximum and minimum count of books issued in each department, maximum and minimum count of books issued considering the entire library, extensively used and less used book in each department, extensively used and less used book in the entire library and percentage ratio of total books in the library to number of books issued to the users. These data will provide a usage pattern that helps in tracking individual library material.

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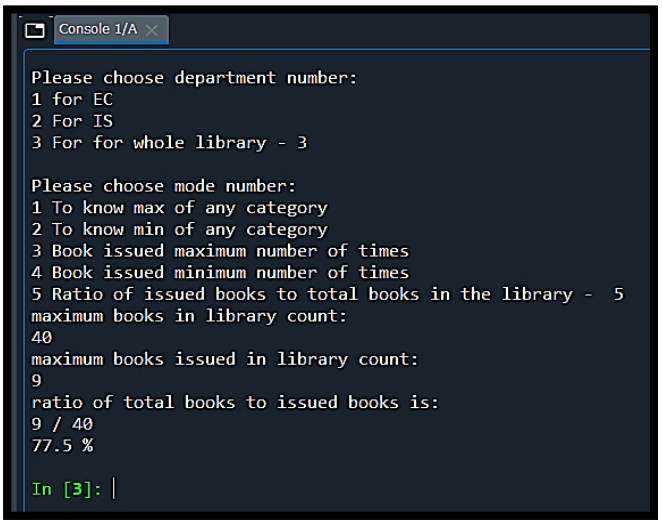

Fig 4: Analytical information of the library

Besides the Desktop app and analytical information, data sets were being created in the CSV formats and made available to the classification algorithms. The decision tree classification algorithm is an algorithm in machine learning by which one can break large datasets in to small subsets and help us make analysis on the same. In the proposed model, decision tree is used to predict to which department a book belongs to in the entire library. In this case, we have obtained results and classified various books into two departmental categories for example, namely EC and IS departments. Below figure 5 depicts the classification between the two specified departments (IS and EC).

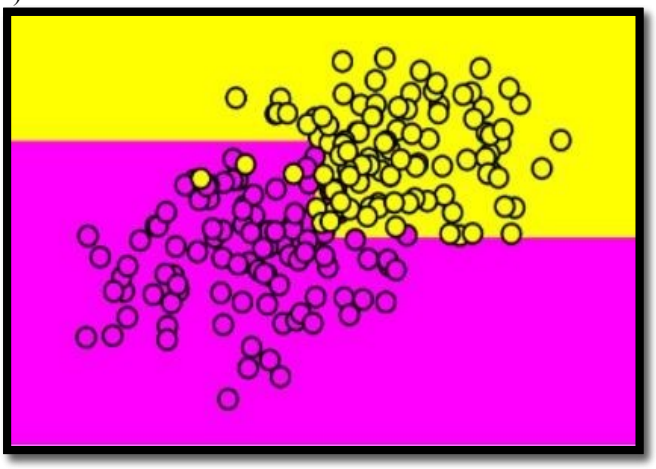

Figure 5: Visual view of the separating border between the two categories of department IS and EC.

The above separator plot was obtained on running the decision tree algorithm. In the above figure 5, Yellow color depicts the Information science books and purple color depicts the Electronics and communication books respectively. This is done with the help of Sklearn decision tree and max_depth parameter that limits the depth of the tree. Thus, we obtain the visual and clear-cut separation of books between the two departments.

Another machine algorithm named Regression is being used to analyze the pattern and number of books being discarded from time to time. This also predicts the future count of books that needs to be discarded.
Figure 6 shows the regression plot obtained after running the regression algorithm. Thus, gives an approximate idea on the count of books that will be discarded in the next two years.

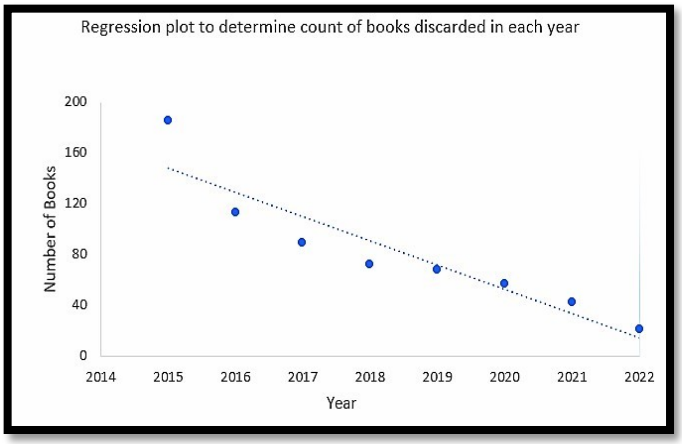

Figure 6: Regression plot obtained along with prediction for the upcoming two years.

C. Creating an Interface Using VNC Viewer

VNC is a graphical desktop that permits system sharing to distantly control the desktop interface of one PC from another PC or mobile device. VNC viewer transmits the console, mouse, or contact occasions to the VNC worker, and further more gets updates to the screen consequently. For the functionality of the system, the Raspberry Pi, PC, or system must be associated with a similar organization interface. This progression is profoundly significant for the best possible functionality of the VNC viewer. Connection with the hostname or address of the organization must be built up. Confirmation of the user by entering a valid user ID and password is performed. The Raspberry Pi screen is open on your system or PC through which one can operate the tasks wanted to be performed. Dump and run the python code in the interface made between Raspberry Pi and the system for the working of the equipment. The hardware modules are at last prepared for their work. When the hardware is powered up and prepared, the RFID cards of books and student RFID cards must be set against the reader for the scanning process to occur. Therefore, tasks such as the issue of books, the return of books, search for books can be performed effectively without the intervention of the Liberian. Once the RFID cards have been scanned by the reader, it is shown on the screen of the Desktop app developed utilizing the PyQt GUI (Graphical User Interface).

D. Tasks Performed by the System

- The Issue Process: When the user wants to get his book issued, he can complete it with no manual intervention. He must zap his RFID card against the RFID reader and it consequently opens his/her account activity page (updated information base).
He should then place the chosen books to be issued, individually before the RFID reader. For example, if at most 3 books can be given under a student's name as rule, then student will have just 3 books under his authority. The PC will record all this information against his name. At last, a pop-up is shown on the screen indicating the client that the book Issue has been done. The client takes the books for a predetermined time from the library after which he should return the books to the library.

- The Return/Submit Process: When the user wants to return his books, he must zap the book's tag again before the RFID reader and the books are automatically adjusted for the return against the user's name.

- The Book Search /Availability Process: When the user wants to check for the availability of the book, he/she should simply compose accurately the book name or the RFID card number

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known in the space given in the Desktop app's dialogue box. Once entered the details effectively, the site gives a tally of the number details effectively, the site gives a tally of the number of such books accessible in the library. It additionally specifies, whether the required book isn't accessible in the library.

- The Student Registration Process: The students have no access to the issue or return book measure in the event that they have not been duly registered with their Roll numbers on the website made. Subsequently, guaranteeing antitheft of books. The student's need to enlist their names by simply zapping their RFID cards given to every student against the RFID reader and present their details in the website's dialogue box. The database updates accordingly.

- Department wise demand of a book in the library: In case of our project, we have considered two departments i.e., the EC department and IS department. This function prints out the book that is in great demand amongst the other books. This is decided by the parameters such as the number of times the book is being issued and enquired.

- The book that is not in demand in each department: This function gives us an insight on the book that is less used by the users which depends on the parameters like the number of times the book is issued and enquired.

- Overall demand of a book in the entire Library: This function gives the information on which book is more frequently required by the users in the entire library. This function also depends on the number of times a book is issued and enquired.

- The Book which is less required in the entire Library: This function gives the information on which book is less frequently required by the users in the entire library. This function also depends on the number of times a book is issued and enquired.

- The ratio of the total number of books in the library to the number of books issued: It involves computing the percentage ratio of the total number of books in the entire library to the number of books issued to the users of the library. This information gives the accurate count of books that are being dealt with from time to time, thus maintaining a Clean Library Management System.

V. Implementation And Results

This section describes the hardware connections and graphical representation of books in each department and count of books issued department wise. The obtained results help us to visualize the overall working and prototype developed.

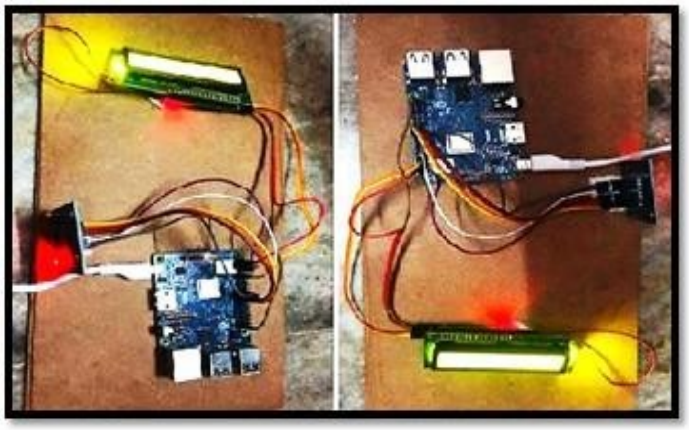

Fig 7: Hardware connections made as per the proposed block diagram

After making the connections as per the block diagram in figure 7, we were able to establish communication between the RFID reader, RFID tags, and Raspberry Pi. The unique Inventory tag numbers embedded onto each book once scanned by the reader is then displayed on the digital display (LCD) for validation purpose. The same inventory number was also displayed in the Desktop app created. The below figure 8 shows the unique number displayed on the LCD screen, the VNC viewer screen and then was transferred to the Desktop app created to perform corresponding actions onto the database.

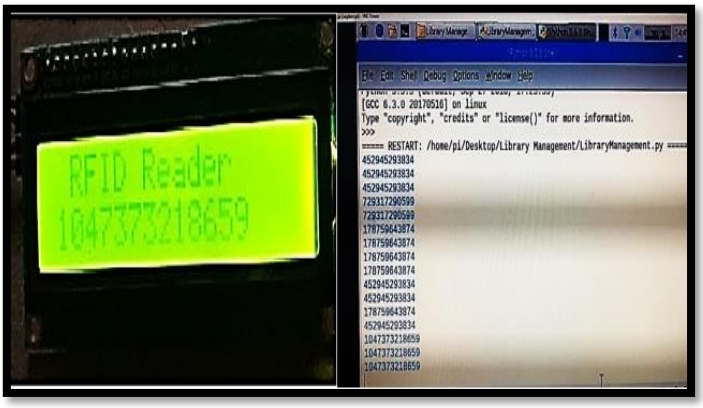

Fig 8: Display of RFID tag numbers scanned by the reader on the LCD screen and VNC viewer

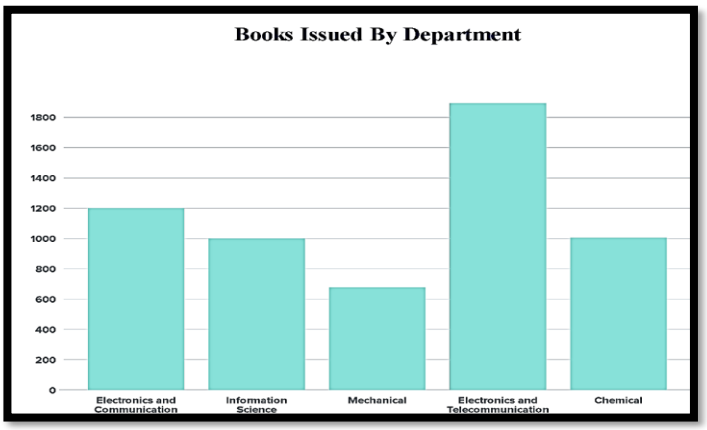

Fig 9: Graphical analysis of books issued department wise

The figure 9 depicts the department wise book issued in the library. This exact count obtained helps us to track and deal with the books without getting lost.

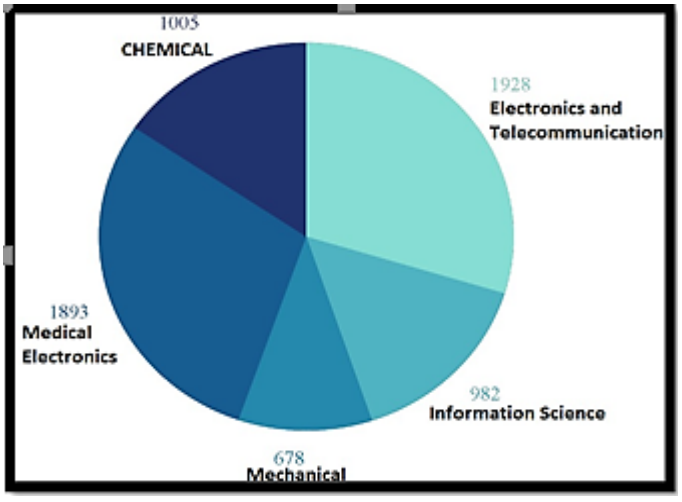

Fig 10: Total number of books in each department.

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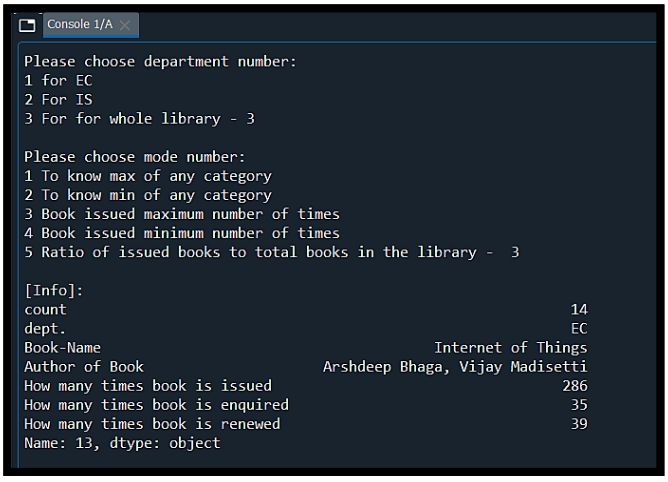
The above figure 10 depicts total number of books in each department. This above feature helps to track the books and gives us an overall idea on the total book contribution by each department.



```
Console 1/A

Please choose department number:
1 for EC
2 For IS
3 For for whole library - 3

Please choose mode number:
1 To know max of any category
2 To know min of any category
3 Book issued maximum number of times
4 Book issued minimum number of times
5 Ratio of issued books to total books in the library -  3

[Info]:
count                                                    14
dept.                                                    EC
Book-Name                              Internet of Things
Author of Book                 Arshdeep Bhaga, Vijay Madisetti
How many times book is issued                           286
How many times book is enquired                          35
How many times book is renewed                           39
Name: 13, dtype: object
```

Fig 11: Display of output obtained after running python code on Spyder

The above output obtained can be explained in the following steps:

- To find extensively used or less used book, run the python code developed on the Spyder platform.
- Enter the respective department or whole library from the drop down of the Desktop app to know the information on maximum or minimum number of times book is issued, enquired and renewed.
- Enter the respective department or whole library from the drop down of the Desktop app to find the percentage ratio of total books to the issued books in the library.

VI. CONCLUSIONS

RFID Technology isn't just emerging, yet additionally more compelling, advantageous with cost-proficient innovation in library management and security concern. Such technology reduces human involvement and thus the errors caused unknowingly and also huge spare in labor costs and client administrations.

In library management some of the implications include, book-tracking, fine calculation and tracing, thus prevents from theft, and proper timely return/response from the users of the library assets, along with easy book search and information availability. In addition, applying machine learning concepts to library management helps classify books, and other analytics relating to predictions and classification. Besides classification and prediction, other statistical methods will help in the library maintenance, usage patterns and inventory control.

The proposed intelligent system includes all the above-mentioned activities along with web interface to monitor the user activities. Implementations are carried out by using PyQt GUI, along with prototype hardware module using Raspberry pi. The system is verified for all the functionalities. Results are verified for the analytics and machine learning performance, seen with respect to predictions and classification.

With all the above-mentioned advantages with RFID Technology along with applied machine learning concepts has proven to be an intelligent way of maintaining, tracking, predicting the library assets in a successful and organized manner with added security enhancements.

VII. FUTURE SCOPE

The Library Management system can be further extended by implementing a robot for book sorting and handling. This can reduce errors, tiresome long work hours and completely automate the libraries. Details like book accession number and shelf number are provided by scanning the RFID card which facilitates shelving of books.

Heat maps can also be provided in the case of multiple libraries in an institution for ease in the maintenance of books across different libraries.

In addition, an ultra-high frequency RFID technology can be deployed for large libraries that have more than one floor. An institution comprising more than two libraries can be made to communicate with each other regarding the library information.

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